School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi

CS406: Mid-semester Examination
25th January 2023

Max. marks: 20

Time: 1 hr

Q 1. Find the real root of the equation $3x = \cos x + 1$, using Newton-Raphson method. 4

Q 2. Solve the following system of equations using the Gauss-Jordan method 4

$$x + y + z = 1$$
$$4x + 3y - z = 6$$
$$3x + 5y + 3z = 4$$

Q 3. Find the smallest positive root of $x^2 - 5x + 1 = 0$, correct to four decimal places 4

Q 4. Test the consistency of the following system of equations

$$x + 10y - z = 3$$
$$2x + 3y + 20z = 7$$
$$9x + 22y + 79z = 45.$$

4

Q 5. Using Lagrange interpolation, find the unique polynomial $P(x)$ of degree 2 or less such that $P(1) = 1$, $P(3) = 27$, $P(4) = 64$. 4

School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi
Mid-semester Examination
CS 406 - Numerical Methods
MCA 1st Semester
8th October 2024

Time: 2 hours Total Marks: 20

Answer all questions.

Q 1. Find the interval in which the positive roots of the equation $x - e^{-x} = 0$ lie. Find the smallest positive roots of the equation using Regula-Falsi method. 5

Q 2. Find the solution of the system of equations

$$4x_1 + x_2 + x_3 = 4$$

$$x_1 + 4x_2 - 2x_3 = 4$$

$$3x_1 + 2x_2 - 4x_3 = 6$$

using the Gauss elimination method. 5

Q 3. Using Lagrange's formula, find the interpolating polynomial P(x) of degree 2 or less that takes the following values. 10

$$x_0 = 1, P(x_0) = 1$$

$$x_1 = 3, P(x_1) = 27$$

$$x_2 = 4, P(x_2) = 64$$

School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi
Mid-semester Examination- II

CS 406 - Numerical Methods
MCA 1st Semester
27 November 2024

Time: 2 hrs

Max. Marks: 20

Q 1. Use Newton's forward difference formula to find the unique polynomial P(x) such that P(1) = 1, P(2) = 0, P(3) = 27 and P(4) = 64 (5)

Q 2. Compute the following integral using Trapezoidal rule for p = 1, and number of points = 3 (5)

$$I_p = \int_0^1 \frac{x^p dx}{x^3 + 10}$$

Q 3. Compute approximation to y(0.6) (4 decimals) of the IVP (10)

$$y' = x(x+y) - 2$$
$$y(0) = 2$$

Using Euler's method with step size h = 0.2

School of Computer and Systems Sciences
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New Delhi
End-semester Examination
CS 406 - Numerical Methods
MCA 1st Semester
25th September 2025

Time: 2 hours Total Marks: 20

Answer all questions

Q 1. What is linear independence? Explain how the concept of linear independence is important for solving a system of linear equations. 5

Q 2. Find the inverse of the coefficient matrix of the system of linear equations

$$x + y + z = 1$$

$$4x + 3y - z = 6$$

$$3x + 5y + 3z = 4$$

using the Gauss-Jordan method and then solve the system. 10

Q 3. Given the equation,

$$x^2 + 5 = e^x$$

first, find the interval in which the positive roots of the equation lie and then, using Newton-Raphson method, find the smallest positive roots of the equation. 5

School of Computer and Systems Sciences
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New Delhi
Mid-semester - 2 Examination
Numerical Methods (CS-406)
11th November 2025

Time: 2 hours Total Marks: 20

Answer all questions.

Q 1. For the given the data below, find the polynomial $P(x)$ such that $P(x_i) = y_i \forall i = 0, 1, 2$ and then, find its first derivative, $P'(x)$ (8)

$$x_0 = 0, y_0 = 0$$

$$x_1 = 1, y_1 = 1$$

$$x_2 = 3, y_2 = 0$$

Q 2. Find the approximate value of

$$I = \int_0^1 f(x)dx$$

where

$$f(x) = \frac{1}{1+x}$$

using the trapezium rule with 4 equal subintervals. (8)

Q 3. What is the role of interpolation in animation. Explain in detail. (4)

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School of Computer and Systems Sciences
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New Delhi-110067

Mid Semester Exam 2025-26

Program: MCA (I Semester) **Maximum Marks:** 30
Course: Numerical Methods (CS406) **Exam Date:** 05-10-2026
Time: 60 Minutes **Student Roll No:** .....

Note: Attempt any five questions. Each question carries 6 marks. Show all necessary steps in numerical calculations. Normal and scientific calculators are permitted. Smartphones and other smart devices are not allowed.

Q.1. Answer the following with proper explanation and suitable examples:

a) Explain finite data representation in numerical computation. [2 Marks]

b) Differentiate between fixed-point and floating-point representation with a suitable example. [2 Marks]

c) Explain overflow and underflow in floating-point arithmetic. [2 Marks]

Q.2. The approximate value of a quantity is 24.96, whereas its exact value is 25.00.

a) Calculate the absolute error. [2 Marks]

b) Calculate the relative error. [2 Marks]

c) Calculate the percentage error. [2 Marks]

Q.3. If $y = x^2 + 3x$, and the measured value of x is $x = 2.0 \pm 0.1$, calculate the approximate absolute error, relative error, and percentage error in y using the propagation of error formula. [6 Marks]

Q.4. Consider the equation $f(x) = x^3 - x - 2 = 0$.

a) Verify that the equation has a root in the interval [1, 2]. [2 Marks]

b) Apply the Bisection Method and perform four iterations to obtain an approximate value of the root. [4 Marks]

Q.5. Use the Newton–Raphson Method to find a root of $f(x) = x^3 - 2x - 5 = 0$, taking the initial approximation as $x_0 = 2$. Perform three iterations and obtain the approximate value of the root. [6 Marks]

Q.6. Use the Secant Method to find a root of $f(x) = x^3 - 2x - 5 = 0$ using the initial approximations $x_0 = 2$ and $x_1 = 3$. Perform three iterations and obtain the approximate value of the root. [6 Marks]

Q.7. Attempt either (a) or (b):

a) Use the Regula Falsi Method to find a root of $f(x) = x^3 - x - 1 = 0$ in the interval [1, 2]. Perform three iterations. [6 Marks]

OR

b) Explain the Regula Falsi Method and derive its iteration formula. Using a suitable numerical example, demonstrate the first two iterations. [6 Marks]

Q.8. Answer the following:

a) What is meant by a root of a nonlinear equation? [1 Mark]

b) State the condition required for applying the Bisection Method. [1 Mark]

c) Write the iteration formula for the Secant Method. [2 Marks]

d) Write the iteration formula for the Newton–Raphson Method. [2 Marks]

— End of Question Paper —